

WAVE MOTION

Wave motion can be thought of as the transport of energy and momentum from one point in space to another without transport of matter. For example: Water waves, sound waves, light waves and radio waves. On the basis of medium, there are two types of wave motion.

i) **Mechanical wave motion:** This sort of wave motion is possible only in media (solid, liquid or gas) which possess inertia as well as elasticity. Water waves and sound waves are example of this type of motion and are, therefore, referred to as mechanical waves.

ii) **Electromagnetic (or non-mechanical) wave motion:** No material medium is necessary for the propagation of this sort of wave motion. Light and radio waves which can travel through empty space belong to this category and are, therefore, referred to as electromagnetic waves.

Types of wave motion:

Generally, there are two distinct types of wave motion.

i) **Transverse wave motion:** In this type of wave motion, the particles of the medium oscillate up and down about their mean or equilibrium position in a direction at right angles to the direction of propagation of the wave motion itself. This form of wave motion, therefore, travels in the form crests and troughs. A crest and an adjoining trough make up one wave or pulse. This type of wave motion is possible in media which possess elasticity of shape or rigidity i.e, in solids & liquids. It is not possible in a gaseous medium. Example: waves travelling along a stretched string or water waves.

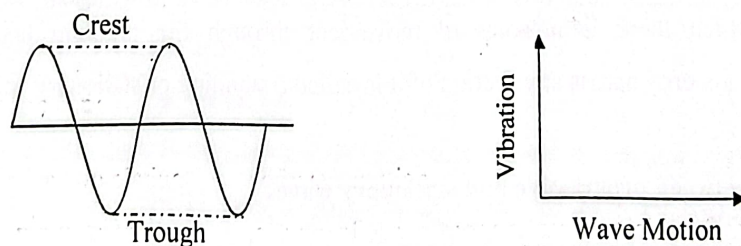


Fig. 1

ii) **Longitudinal wave motion:** In this type of wave motion, the particles of the medium oscillate to and fro about their mean or equilibrium position along the direction of propagation of the wave motion itself. The wave motion, therefore, travels in the form of compressions and rarefactions. One compression and an adjoining rarefaction constitute one wave or pulse. Example: Sound waves.

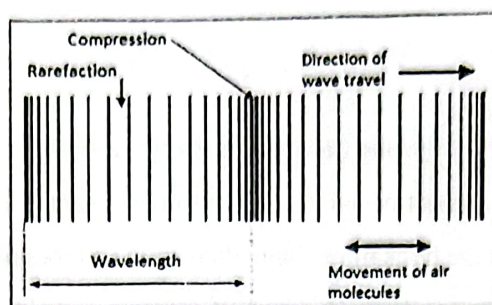


Fig. 2

Production and propagation of wave motion:

The production and propagation of wave motion through a medium, which possesses elasticity and inertia, will now be discussed. No particle of an elastic medium can be disturbed without affecting its immediate neighbour and tending to recover its original position; it first stores up potential energy and then converts it back into kinetic energy. The neighbouring particle which has thus been disturbed and then performs a similar motion, so that each successive particle repeats, in turn, the movements of its predecessor a little later than it and then hands down on to its successor. Thus, a wave motion may be defined as a disturbance or a condition that travels onwards through a medium due to the repeated periodic motion of its particles about their mean or equilibrium positions, each particle repeating the movements of its predecessor a little later than it and handing it on to its successor, so that there is a regular phase difference between one particle and the next. On the basis of propagation, there are two types of wave motion.

✓ A wave motion which progresses onwards through a medium, with energy transferred across every section of it, is called a travelling or a progressive wave. On the other hand, a wave motion in which there is no onward movement through the medium and hence no transference of energy across any section of it is called a standing or stationary wave motion.

Distinguish between progressive and stationary waves

Progressive Wave	Stationary Wave
1) The mode of vibration of each particle of the medium is the same and is handed from one particle to other, so that there is an onward propagation of the wave through the medium.	1) The mode of vibration remains confined to limited space in which it is produced.

i.e., Phase difference $\phi = \frac{2\pi}{\lambda} x$

Hence, the displacement at P is given by

$$\begin{aligned} y &= a \sin (w t - \phi) \\ &= a \sin \left(w t - \frac{2\pi}{\lambda} x \right) \\ &= a \sin \left(2\pi f t - \frac{2\pi x}{\lambda} \right) \\ &= a \sin \left(2\pi \frac{v}{\lambda} t - \frac{2\pi x}{\lambda} \right) \\ &= a \sin \frac{2\pi}{\lambda} (vt - x) \end{aligned}$$

This is the equation of a wave motion denotes a wave moving to right. If the wave is moving to the left then

$$y = a \sin \frac{2\pi}{\lambda} (vt + x)$$

Other forms:

i) $y = a \sin (wt - kx)$

ii) $y = a \sin 2\pi \left(vt - \frac{x}{\lambda} \right)$

iii) $y = a \sin w \left(t - \frac{x}{v} \right)$

iv) $y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$

v) $y = a \sin 2\pi \left(nt - \frac{x}{\lambda} \right)$

Differential equation of a plane progressive wave:

We know the equation of a progressive wave

$$y = a \sin \frac{2\pi}{\lambda} (vt - x) \dots\dots\dots (i)$$

where, a is amplitude, v is the wave velocity and λ is the wave length

Differentiating this equation w. r. t. t , we get

$$\text{Particle velocity, } \bar{U} = \frac{dy}{dt} = \frac{2\pi v}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x) \dots\dots\dots (ii)$$

Differentiating again w. r. t. x , the slope of the displacement curve

$$\frac{dy}{dx} = -\frac{2\pi}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x) \dots\dots\dots (iii)$$

From equation (ii) and (iii)

$$\bar{U} = \frac{dy}{dt} = -v \frac{dy}{dx}$$

i.e., Particle velocity = - (wave velocity) × (slope of the displacement curve)

Differentiating equation (i) w.r.t. t and x twice

$$\therefore \frac{d^2 y}{dt^2} = -\frac{4\pi^2 v^2}{\lambda^2} a \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\therefore \text{acceleration} = -\frac{4\pi^2 v^2}{\lambda^2} y \dots \dots \dots \text{(iv)}$$

The -ve sign shows that acceleration is directed towards the mean position.

$$\begin{aligned} \therefore \frac{d^2 y}{dx^2} &= -\frac{4\pi^2}{\lambda^2} a \sin \frac{2\pi}{\lambda} (vt - x) \\ &= -\frac{4\pi^2}{\lambda^2} y \dots \dots \dots \text{(v)} \end{aligned}$$

So, $\frac{d^2 y}{dt^2} = v^2 \frac{d^2 y}{dx^2} \dots \dots \dots \text{(vi)}$

This is differential equation of a plane wave.

Energy of a plane progressive wave:

Now if the density of the medium is ρ , its kinetic energy per unit volume,

$$\begin{aligned} K.E. &= \frac{1}{2} \rho \bar{U}^2 \\ &= \frac{1}{2} \rho \left[\frac{2\pi v}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x) \right]^2 \\ &= \frac{2\pi^2 \rho v^2 a^2}{\lambda^2} \cos^2 \left[\frac{2\pi}{\lambda} (vt - x) \right] \dots \dots \dots \text{(vii)} \end{aligned}$$

To move the particle from its mean position to a distance y , work has to be done against acceleration. Work done per unit volume for a small displacement dy ,

$$= F dy = \rho \frac{d^2 y}{dt^2} dy = \rho \frac{4\pi^2 v^2}{\lambda^2} y dy$$

Then, potential energy per unit volume,

$$P.E. = \frac{4\pi^2 \rho v^2}{\lambda^2} \int_0^y y dy = \frac{4\pi^2 \rho v^2}{\lambda^2} \cdot \frac{y^2}{2} = \frac{2\pi^2 \rho v^2 y^2}{\lambda^2}$$

Maximum value of particle velocity $U_{\max} = \frac{2\pi a v}{\lambda}$
 # Maximum value of acceleration when $y = a$ $f_{\max} = -\left(\frac{4\pi^2 v^2}{\lambda^2}\right) a$
 $\therefore$ Maximum Particle velocity = $\frac{2\pi a}{\lambda} \times \text{Wave Velocity}$

$$\therefore P.E. = \frac{2\pi^2 \rho v^2 a^2}{\lambda^2} \sin^2 \left[\frac{2\pi}{\lambda} (vt - x) \right] \dots\dots\dots (viii)$$

Total energy per unit volume i.e., energy density of the medium

$$E = P.E. + K.E.$$

$$= \frac{2\pi^2 \rho v^2 a^2}{\lambda^2} \left[\sin^2 \frac{2\pi}{\lambda} (vt - x) + \cos^2 \frac{2\pi}{\lambda} (vt - x) \right]$$

$$\therefore E = \frac{2\pi^2 \rho v^2 a^2}{\lambda^2} \dots\dots\dots (ix)$$

But $v = n\lambda$

$$\therefore E = 2\pi^2 \rho n^2 a^2$$

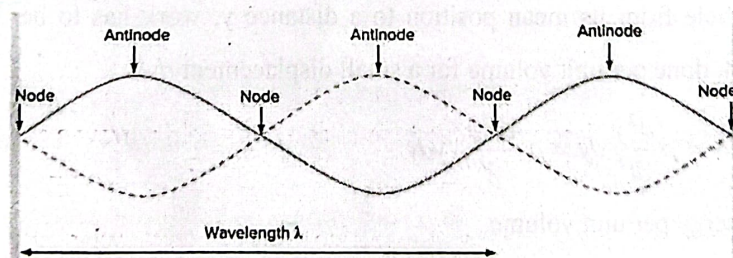
The average kinetic energy per unit volume and the average potential energy per unit volume are equal and each is equal to half of the total energy per unit volume.

$$\text{Average K.E. per unit volume} = \pi^2 \rho n^2 a^2$$

$$\text{Average P.E. per unit volume} = \pi^2 \rho n^2 a^2$$

Stationary wave:

Stationary wave is a wave which oscillates in time but whose peak amplitude profile does not move in space. This wave is produced when two waves of same period, wavelength, amplitude and velocity travelling in opposite direction along the same straight line superimpose upon each other. Such waves are confined in the region of the medium where they are formed a non-propagatory in character. The peak amplitude of the wave oscillations at any point in space is constant with time and the oscillations at different points throughout the wave are in phase. The locations at which the absolute value of the amplitude is minimum are called nodes and the locations where the absolute value of the amplitude is maximum are called antinodes.



Analytical treatment of stationary waves:

(a) Reflection occurs at free boundary or open end or string free

In this case the displacement of the particle continues in the same direction as before with no phase changes. Let the wave trains be represented by

$$y_1 = a \sin \frac{2\pi}{\lambda} (vt - x) \quad \text{Incident wave}$$

and
$$y_2 = a \sin \frac{2\pi}{\lambda} (vt + x) \quad \text{Reflected wave}$$

The resultant displacement, y of the particle as a consequence of superposition is

$$\begin{aligned} y &= y_1 + y_2 = a \sin \frac{2\pi}{\lambda} (vt - x) + a \sin \frac{2\pi}{\lambda} (vt + x) \\ &= 2a \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi vt}{\lambda} \dots\dots\dots(1) \end{aligned}$$



Fig.: Reflection occurs at free boundary.

Therefore, the resultant wave is simple harmonic with same time period and amplitude,

$$A = 2a \cos \frac{2\pi x}{\lambda}$$

Case I: The displacement is zero or minimum at points

when
$$\cos \frac{2\pi x}{\lambda} = 0 = \cos(2n+1) \frac{\pi}{2}$$

or,
$$x = (2n+1) \frac{\lambda}{4} \quad \text{where, } n = 0, 1, 2, 3, \dots \text{etc.}$$

or,
$$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4} \quad \text{etc.} \quad \text{(nodes)}$$

Hence, the amplitude of the resultant vibration will be zero and the displacements at these points are also zero i.e., $y = 0$. These points are called nodes. They are therefore points in the medium which are permanently at rest. At the nodes, the strain is maximum.

Case II: The displacement is maximum at points

when
$$\cos \frac{2\pi x}{\lambda} = \pm 1 = \cos n\pi$$

or,
$$x = \frac{n\lambda}{2}$$

or,
$$x = 0, \frac{\lambda}{2}, \lambda, \quad \text{etc.} \quad \text{(antinodes)}$$

Hence, the amplitude of the resultant vibration, $A = 2a \cos \frac{2\pi x}{\lambda}$ has maximum value and the displacements at these points are maximum i. e.,

$$y = 2a \sin \frac{2\pi vt}{\lambda}$$

These points are called antinodes which are separated by a distance of half a wave length. At the antinodes, the strain is minimum.

(b) Reflection occurs at a fixed boundary

In this case, the displacement of the particle at a fixed boundary is exactly cancelled by an equal and opposite displacement due to reflection. i.e., displacement under goes a phase change π , but not the strain and pressure.

Let the equation of a wave in $+x$ direction

$$y_1 = a \sin \frac{2\pi}{\lambda} (vt - x)$$

The equation of the reflected wave

$$y_2 = -a \sin \frac{2\pi}{\lambda} (vt + x)$$

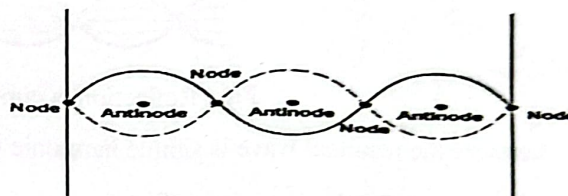


Fig.: Reflection occurs at fixed boundary.

$$\therefore y = y_1 + y_2 = -2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda}$$

Therefore, the resultant wave is simple harmonic with same time period and the amplitude,

$$A = 2a \sin \frac{2\pi x}{\lambda}$$

Case 1: The displacement is zero at points

$$\text{when } \sin \frac{2\pi x}{\lambda} = 0 = \sin n\pi$$

$$\text{or, } x = \frac{n\lambda}{2} \quad \text{where, } (n = 0, 1, 2, 3, \dots \text{etc.})$$

$$\text{or, } x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \dots \text{etc.} \dots \dots \dots (\text{nodes})$$

Hence, the displacement of the particles is zero at these points and these points are called nodes.

Case 11: The displacement of the particle is maximum

when $\sin \frac{2\pi x}{\lambda} = \pm 1 = \sin(2n+1)\frac{\pi}{2}$

or, $x = (2n+1)\frac{\lambda}{4}$

or, $x = \lambda/4, 3\lambda/4, 5\lambda/4, \dots \text{---etc.} \dots \dots \dots (\text{antinodes})$

Hence, these points are called antinodes.